

PSG COLLEGE OF TECHNOLOGY, COIMBATORE – 641 004

DEPARTMENT OF COMPUTER APPLICATIONS
Academic Year 2024-2025 23MX18 WEB APPLICATION
DEVELOPMENT

StoreFlow–A Smart ERP Solution for Chain Store Management

I. Team Detail

TEAM NO. & NAME	ROLL NO.	NAME OF THE STUDENT	GUIDE NAME
5.AKATSUKI	25MX304	ARAVINDH KANNAN M S	DR SUBATHRA M
	25MX330	MONISH P	
	25MX335	NAVEEN PRANAB T	

II. Title of the Application

StoreFlow – A Smart ERP Solution for Chain Store Management

III. Overview of the Project

StoreFlow is a simple ERP web application built using the MERN stack (MongoDB, Express.js, React.js, Node.js). It helps small and medium chain stores manage daily business tasks easily from one place.

The system has modules for inventory tracking, order management, financial reports, and appointment scheduling.

A clear dashboard shows revenue, expenses, and profit with visual charts. The calendar helps track store activities like orders, meetings, and restocks.

StoreFlow helps multiple branches share information in real time, making store operations smoother and more organized.

IV. Existing System

Most small businesses still use manual methods or spreadsheets for managing sales, stock, and expenses.

This makes it hard to track performance, causes mistakes, and wastes time.

Big ERP systems exist, but they are too expensive and complex for small stores.

Hence, a user-friendly, low-cost ERP is needed for better control and efficiency.

VI. Objectives of the Proposed Application

- Build an easy-to-use ERP for small chain stores.
- Combine all store activities like stock, sales, and expenses in one system.
- Show real-time financial updates (revenue, expenses, profit).
- Add a calendar for appointments and events.
- Provide live order tracking and stock management.
- Improve decision-making with a clear dashboard.
- Make the system secure and scalable using MERN.

VII. Scope / Use

StoreFlow can be used by shop owners, vendors, and small business managers. It helps them manage store operations without needing technical skills. The system can also be expanded to include multiple branches or local communities, making it useful for any growing business chain.

VIII. Technology to be used

Category	Technology
Frontend	React.js, HTML, CSS, JavaScript
Backend	Node.js, Express.js
Database	MongoDB
Authentication	JWT (JSON Web Token)
Deployment	Netlify/Vercel (Frontend), Render/Railway (Backend)

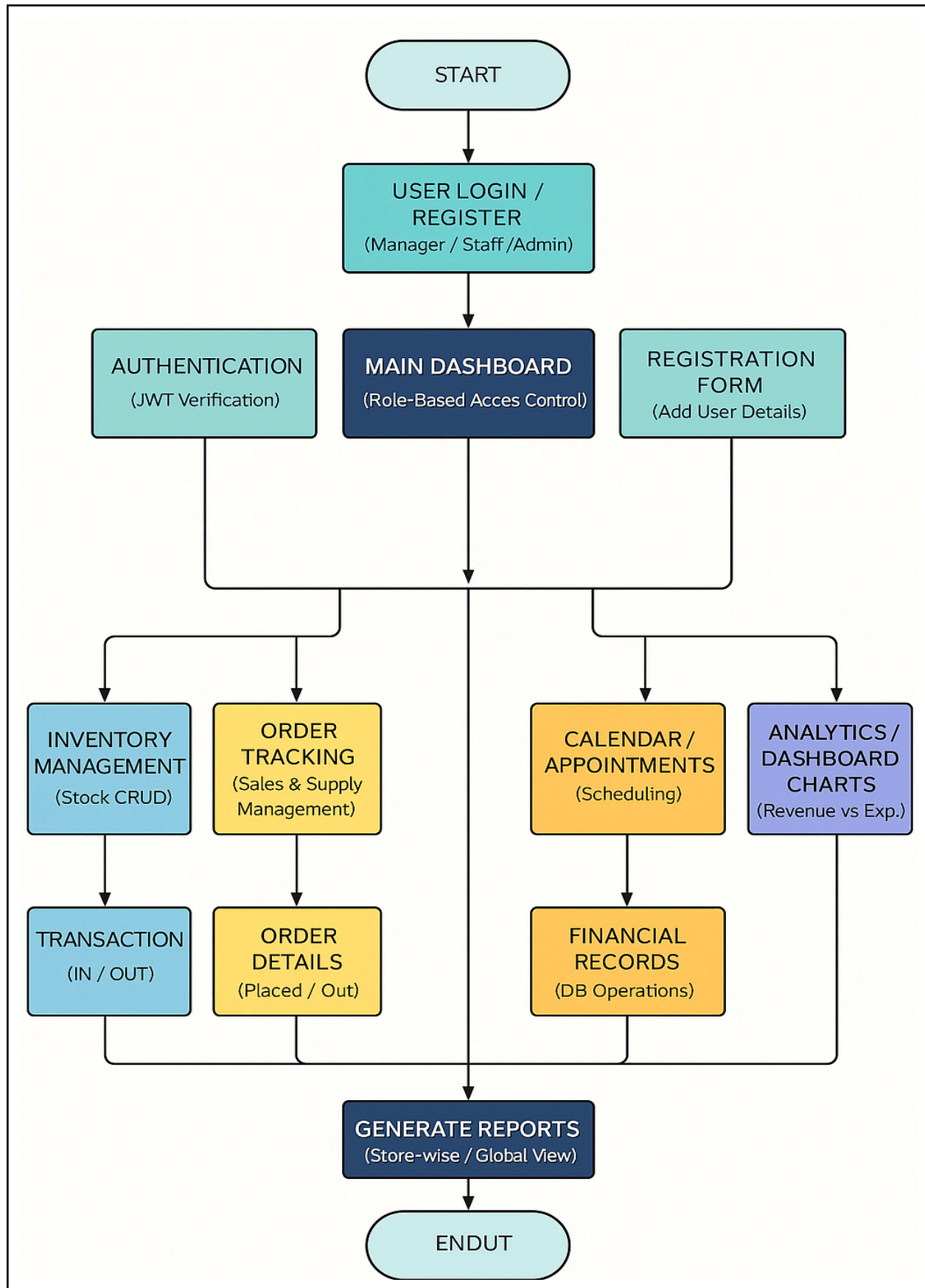
IX. Functional Requirements of the Application

- Login & Registration: Secure login for store admins and staff.
- Dashboard: Shows revenue, expenses, profit, and charts.
- Inventory Management: Add, edit, and track stock levels.
- Order Tracking: Manage customer orders across branches.
- Calendar: Schedule meetings and events.
- Reports: Generate daily/weekly financial summaries.

X. Non-Functional Requirements of the Application

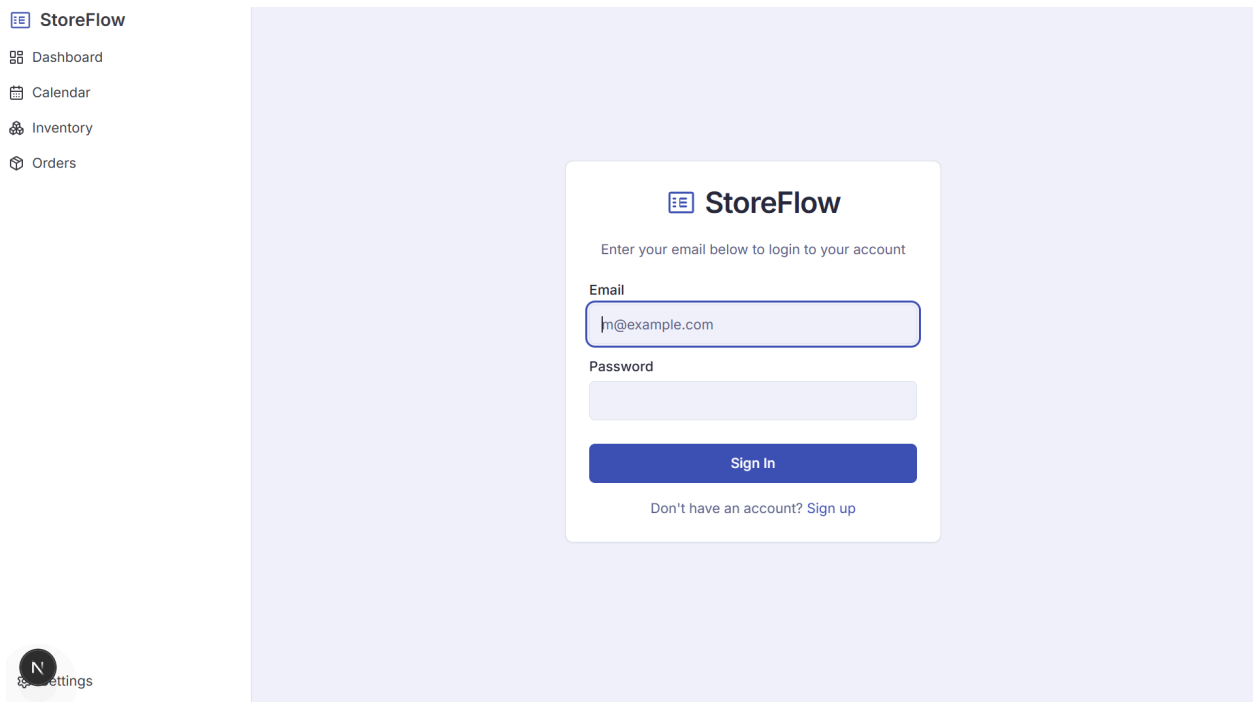
- Performance: Quick data access and updates.
- Scalability: Supports multiple stores.
- Usability: Simple interface for non-technical users.
- Security: JWT authentication and secure APIs.
- Responsiveness: Works well on mobile and desktop.

XI. Flow diagram of the Application

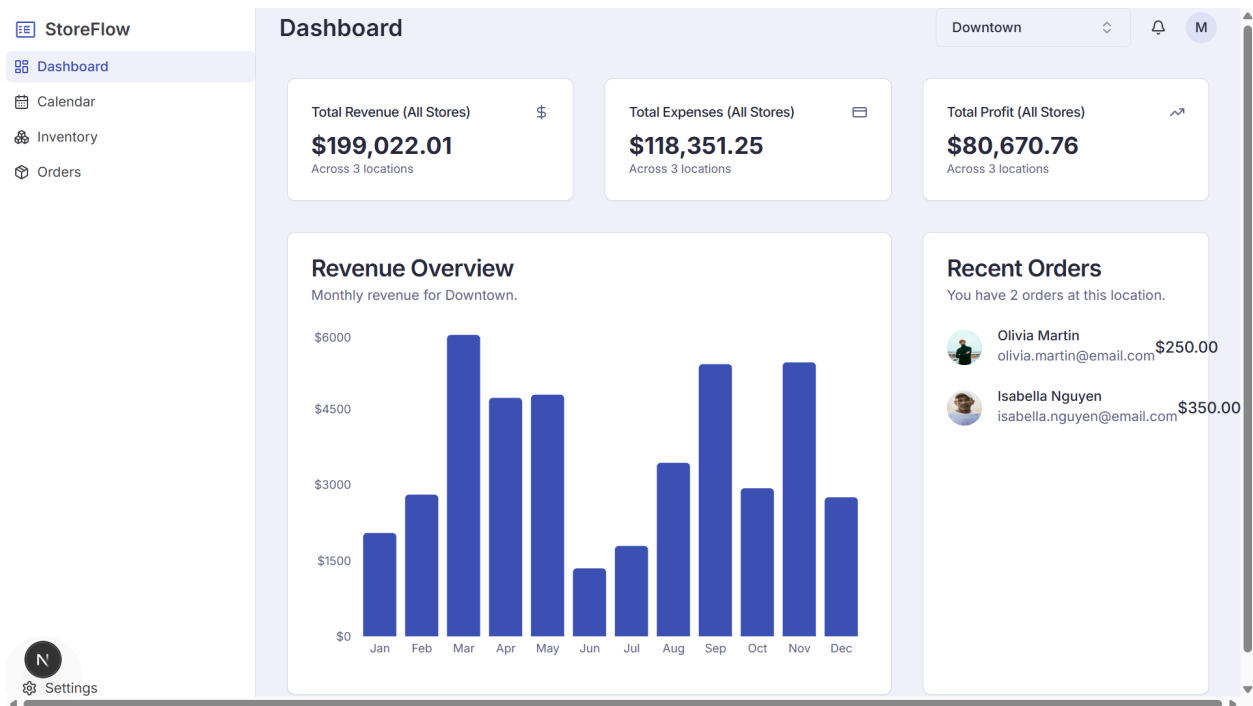


XII. Interface/prototype of the proposed Application

LOGIN PAGE:



DASHBOARD:



CALENDER:

StoreFlow

Dashboard

Calendar

Inventory

Orders

Calendar

Downtown

M

Upcoming Appointments

For Downtown location.

New Appointment

Service A

Alice Johnson

Friday, November 7

11:00 AM - 12:00 PM

With: Mike Brown

N

Settings

INVENTORY:

StoreFlow

Dashboard

Calendar

Inventory

Orders

Inventory

Downtown

M

Inventory for Downtown

Track and manage your product stock.

Product	SKU	Status	Stock
Product A	P-001	In Stock	150
Product B	P-002	Low Stock	18
Product C	P-003	Out of Stock	0

N

Settings

ORDERS:

- StoreFlow
- Dashboard
- Calendar
- Inventory
- Orders

Settings

Orders

Downtown

M

Orders for Downtown

Manage and track customer orders.

Customer	Date	Status	Amount
Olivia Martin	11/7/2025	Fulfilled	\$250.00
Isabella Nguyen	11/6/2025	Shipped	\$350.00

Signature of the students

Signature of the Guide